

9.2.3 Application

During application, the input pattern is compared with the exemplar/code vector in terms of the distance between them. The cluster unit whose code vector is at a least distance from the input pattern is the winner. The input pattern belongs to the corresponding cluster.

The learning and application process of Kohonen's self-organizing map are illustrated in Example 9.2 and Problem 9.1 in the Solved Problems section respectively.

Example 9.2 (Learning by Kohonen's self-organizing map)

Suppose there are four patterns $s_1 = [1, 0, 0]$, $s_2 = [0, 0, 1]$, $s_3 = [1, 1, 0]$ and $s_4 = [0, 1, 1]$ to be clustered into two clusters. The target SOM, as shown in Fig. 9.9, consists of three input units and two output units.

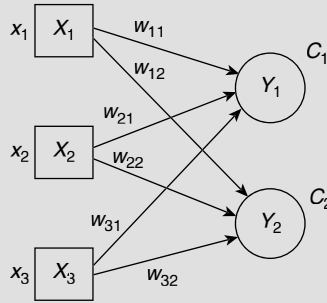


Fig. 9.9. Target SOM network

The exemplar, or code-vector, for the clusters C_1 and C_2 , represented by the cluster units Y_1 and Y_2 respectively, are given by the 1st and the 2nd column of the weight matrix

$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{bmatrix}$$

Hence $W_{\cdot 1} = [w_{11}, w_{21}, w_{31}]^T$ and $W_{\cdot 2} = [w_{12}, w_{22}, w_{32}]^T$ are the code-vectors for the clusters C_1 and C_2 respectively. Now let us denote the input vectors as $s_1 = [1, 0, 0]$, $s_2 = [0, 0, 1]$, $s_3 = [1, 1, 0]$ and $s_4 = [0, 1, 1]$. The successive steps of the learning process are described below.

Step 0. Initialize

- (i) Weight matrix W is randomly initialized as

$$W = \begin{bmatrix} .5 & .3 \\ .8 & .5 \\ .4 & .3 \end{bmatrix}$$

- (ii) Topological neighbourhood is initialized to $R = 0$, because there are only two cluster units.
- (iii) Learning rate $\eta = 0.8$. It will be geometrically decreased with a factor of 0.5 after each epoch of training.

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Step 1. **While** stopping condition is not satisfied

Do Steps 2-8.

When the weight adjustment $\Delta w_{ij} < 0.01$ the net is assumed to have converged. Initially the stopping condition is obviously false. Hence we continue to Step 2.

Step 2. **For** each training vector $\mathbf{s}_i$ **Do** Step 3 to 5.

We start training with the first vector $\mathbf{s}_1 = [1, 0, 0]$.

Step 3. **For** each exemplar /code vector $\mathbf{w}_{*j}$
Do find the distance $D(j)$ between $\mathbf{s}_i$ and $\mathbf{w}_{*j}$.

$$D(j) = \sum_i (x_i - w_{ij})^2$$

$$\begin{aligned} D(1) &= \text{distance between } \mathbf{s}_1 \text{ and } \mathbf{w}_{*1} \\ &= (1 - 0.5)^2 + (0 - 0.8)^2 + (0 - 0.4)^2 \\ &= 0.25 + 0.64 + 0.16 \\ &= 1.05 \end{aligned}$$

$$\begin{aligned} D(2) &= \text{distance between } \mathbf{s}_1 \text{ and } \mathbf{w}_{*2} \\ &= (1 - 0.3)^2 + (0 - 0.5)^2 + (0 - 0.3)^2 \\ &= 0.49 + 0.25 + 0.09 \\ &= 0.83 \end{aligned}$$

Step 4. Find index J so that $D(J)$ is minimum.

Since, $D(2) < D(1)$, $\mathbf{s}_1$ is closer to C_2 , and Y_2 is the winner. Therefore, the code vector $\mathbf{w}_{*2}$ is to be adjusted.

Step 5. Update the weights of all units within the specified neighbourhood of J .

For all J , such that $J - R \leq j \leq J + R$ **Do**

For all $i = 1$ to m **Do**

$$w_{ij}(\text{new}) = w_{ij}(\text{old}) + \eta \times [x_i - w_{ij}(\text{old})]$$

Since $R = 0$, we need to adjust the weight vector of the winner only.

$$\begin{aligned} w_{12}(\text{new}) &= w_{12}(\text{old}) + \eta \times (x_1 - w_{12}(\text{old})) \\ &= 0.3 + 0.8 \times (1 - 0.3) \\ &= 0.3 + 0.8 \times 0.7 \\ &= 0.86 \end{aligned}$$

$$\begin{aligned} w_{22}(\text{new}) &= w_{22}(\text{old}) + \eta \times (x_2 - w_{22}(\text{old})) \\ &= 0.5 + 0.8 \times (0 - 0.5) \\ &= 0.5 - 0.4 \\ &= 0.1 \end{aligned}$$

$$\begin{aligned}
w_{32}(\text{new}) &= w_{32}(\text{old}) + \eta \times (x_3 - w_{32}(\text{old})) \\
&= 0.3 + 0.8 \times (0 - 0.3) \\
&= 0.3 - 0.24 \\
&= 0.06
\end{aligned}$$

Hence, the weight matrix after training with the 1st input vector in the 1st epoch is

$$W = \begin{bmatrix} 0.5 & 0.86 \\ 0.8 & 0.1 \\ 0.4 & 0.06 \end{bmatrix}$$

We now proceed to train the SOM with the second pattern.

Step 2. **For** each training vector $\mathbf{s}_i$ **Do** Step 3 to 5.

After training with the first vector $\mathbf{s}_1 = [1, 0, 0]$, we proceed with the second pattern $\mathbf{s}_2 = [0, 0, 1]$.

Step 3. **For** each exemplar /code vector $\mathbf{w}_{*j}$

Do find the distance $D(j)$ between $\mathbf{s}_i$ and $\mathbf{w}_{*j}$.

$$D(j) = \sum_i (x_i - w_{ij})^2$$

$$\begin{aligned}
D(1) &= \text{distance between } \mathbf{s}_2 \text{ and } W_{*1} \\
&= (0 - 0.5)^2 + (0 - 0.8)^2 + (1 - 0.4)^2 \\
&= 0.25 + 0.64 + 0.36 \\
&= 1.25
\end{aligned}$$

$$\begin{aligned}
D(2) &= \text{distance between } \mathbf{s}_2 \text{ and } W_{*2} \\
&= (0 - 0.86)^2 + (0 - 0.1)^2 + (1 - 0.06)^2 \\
&= 0.74 + 0.01 + 0.88 \\
&= 1.63
\end{aligned}$$

Step 4. Find index J so that $D(J)$ is minimum.

Since, $D(1) < D(2)$, $\mathbf{s}_2$ is closer to C_1 , and Y_1 is the winner. Therefore, the code vector W_{*1} is to be adjusted.

Step 5. Update the weights of all units within the specified neighbourhood of J .

For all J , such that $J - R \leq j \leq J + R$ **Do**
For all $i = 1$ to m **Do**

$$w_{ij}(\text{new}) = w_{ij}(\text{old}) + \eta \times [x_i - w_{ij}(\text{old})]$$

Since $R = 0$, we need to adjust the weight vector of the winner only.

$$\begin{aligned}
w_{11}(\text{new}) &= w_{11}(\text{old}) + \eta \times (x_1 - w_{11}(\text{old})) \\
&= 0.5 + 0.8 \times (0 - 0.5) \\
&= 0.5 - 0.8 \times 0.5 \\
&= 0.1 \\
w_{21}(\text{new}) &= w_{21}(\text{old}) + \eta \times (x_2 - w_{21}(\text{old})) \\
&= 0.8 + 0.8 \times (0 - 0.8) \\
&= 0.8 - 0.64 \\
&= 0.16 \\
w_{31}(\text{new}) &= w_{31}(\text{old}) + \eta \times (x_3 - w_{31}(\text{old})) \\
&= 0.4 + 0.8 \times (1 - 0.4) \\
&= 0.4 + 0.48 \\
&= 0.88
\end{aligned}$$

Hence, the weight matrix W after training with the second vector in the first epoch is

$$W = \begin{bmatrix} 0.1 & 0.86 \\ 0.16 & 0.1 \\ 0.88 & 0.06 \end{bmatrix}$$

In this way learning with the rest of the patterns $s_3 = [1, 1, 0]$ and $s_4 = [0, 1, 1]$ takes place by repeating Steps 3–5. Table 9.2 shows the outline of learning during the first epoch.

Table 9.2. Learning by the SOM during the first epoch

#	Training Patterns $s = (x_1, x_2, x_3)$	Squared Euclidean Distance		Winner	New Code Vectors	
		$D(1)$	$D(2)$		$C_1 (w_{11}, w_{21}, w_{31})$	$C_2 (w_{12}, w_{22}, w_{32})$
0					(0.5, 0.8, 0.4)	(0.3, 0.5, 0.3)
1	$s_1 = (1, 0, 0)$	1.05	0.82	C_2	No change	(0.86, 0.1, 0.06)
2	$s_2 = (0, 0, 1)$	1.25	1.63	C_1	(0.1, 0.16, 0.88)	No change
3	$s_3 = (1, 1, 0)$	2.29	1.71	C_2	No change	(0.97, 0.82, 0.01)
4	$s_4 = (0, 1, 1)$	1.53	1.95	C_1	(0.02, 0.83, 0.98)	No change

So, code-vector W_{*2} gets modified as a result of training with the patterns s_1 and s_3 while training with the patterns s_2 and s_4 results in adjustment of the code-vector W_{*1} .

The weight matrix obtained after the first epoch is

$$W_1 = \begin{bmatrix} 0.02 & 0.97 \\ 0.83 & 0.82 \\ 0.98 & 0.01 \end{bmatrix}$$

On further computation, the weight matrix after the second epoch becomes

$$W_2 = \begin{bmatrix} 0.01 & 0.99 \\ 0.7 & 0.7 \\ 0.99 & 0 \end{bmatrix}$$

The training process converges after 19 epochs when the weight matrix becomes (approximately)

$$W_{19} = \begin{bmatrix} 0 & 1 \\ 0.6 & 0.6 \\ 1 & 0 \end{bmatrix}$$

Fig. shows the SOM obtained for the given clustering problem.

